

Lecture 19: Line Integrals

Background & Motivation

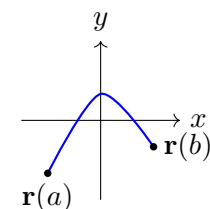
With vector fields defined, we now integrate them along curves. The line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ computes the work done by a force along a path. For conservative fields, the Fundamental Theorem of Line Integrals gives us a shortcut: the integral depends only on the endpoints, not the path.

1. Contour Types

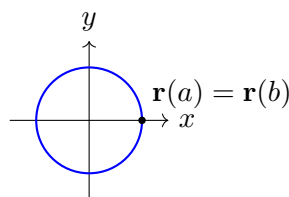
A **contour** (or **curve**) C is a path in $\mathbb{R}^2$ or $\mathbb{R}^3$ given by a parameterization $\mathbf{r}(t)$, $a \leq t \leq b$.

Definition 1: Types of Contours

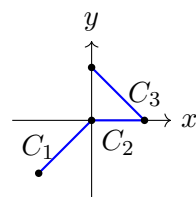
- **Open** – endpoints $\mathbf{r}(a) \neq \mathbf{r}(b)$
- **Closed** – $\mathbf{r}(a) = \mathbf{r}(b)$ (loop); use notation $\oint_C$
- **Simple** – does not cross itself
- **Piecewise smooth** – finitely many smooth pieces joined end-to-end



Open Contour



Closed Contour



Piecewise Contour

Common Parameterizations

- Line segment from A to B – $\mathbf{r}(t) = (1-t)A + tB$, $t \in [0, 1]$
- Circle $x^2 + y^2 = R^2$ – $\mathbf{r}(t) = \langle R \cos t, R \sin t \rangle$, $t \in [0, 2\pi]$
- Graph $y = f(x)$, $a \leq x \leq b$ – $\mathbf{r}(t) = \langle t, f(t) \rangle$, $t \in [a, b]$
- Helix – $\mathbf{r}(t) = \langle a \cos t, a \sin t, bt \rangle$

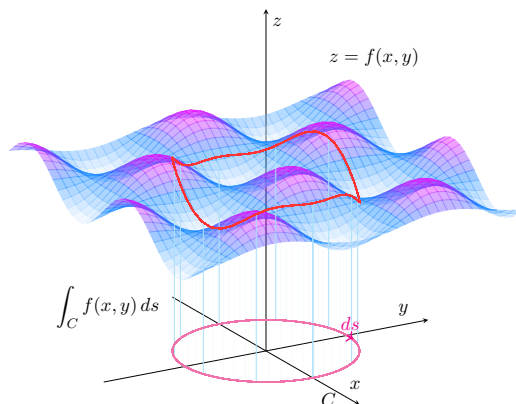
2. Scalar Line Integrals

Definition 2: Scalar Line Integral

For a scalar function f along a curve C parameterized by $\mathbf{r}(t)$:

$$\int_C f \, ds = \underline{\hspace{2cm}}$$

- $ds = \|\mathbf{r}'(t)\| \, dt$ is the **arc length element**
- **Interpretation:** area of the “curtain” under f along C
- Direction does not matter: $\int_C f \, ds = \int_{-C} f \, ds$



Example 1: Scalar Line Integral Along a Line Segment

Evaluate $\int_C (x + 2y) \, ds$ where C is the segment from $(0, 0)$ to $(3, 4)$.

3. Vector Field Line Integrals

Definition 3: Line Integral of a Vector Field

For a vector field $\mathbf{F}$ along a curve C :

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \underline{\hspace{2cm}}$$

- **Interpretation:** $\underline{\hspace{2cm}}$
- Reversing direction **negates** the integral: $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r}$

– Equivalent notation: $\int_C P dx + Q dy + R dz$

Example 2: Work Around a Circle

Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle -y, x \rangle$ along the circle $x^2 + y^2 = 4$ (CCW).

4. Fundamental Theorem of Line Integrals (FTOLI)

Theorem 1: Fundamental Theorem of Line Integrals

If $\mathbf{F} = \nabla f$ (conservative) and C goes from A to B :

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$$

Consequences:

- The integral depends only on _____ – **path independence**
- For any closed curve: $\oint_C \nabla f \cdot d\mathbf{r} = \underline{\hspace{2cm}}$
- If $\oint_C \mathbf{F} \cdot d\mathbf{r} \neq 0$ for some closed C , then $\mathbf{F}$ is **not** conservative

Example 3: Applying FTOLI

Evaluate $\int_C \langle ye^{xy}, xe^{xy} \rangle \cdot d\mathbf{r}$ from $(0,0)$ to $(1,2)$.

Example 4: Piecewise 3D Line Integral

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y^2, z^2, x^2 \rangle$ along the path $(0, 0, 0) \rightarrow (1, 0, 0) \rightarrow (1, 1, 0) \rightarrow (1, 1, 1)$.

Practice Problems

Problem 1. Evaluate $\int_C xy \, ds$ where C is the line segment from $(0, 0)$ to $(1, 1)$.

Problem 2. Evaluate $\int_C \langle y^2, 2xy \rangle \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 1)$ along $y = x^2$.

Problem 3. Evaluate $\oint_C \langle x^2, y^2 \rangle \cdot d\mathbf{r}$ where C is the unit circle. (*Hint: is the field conservative?*)

Problem 4. Evaluate $\int_C \langle 2xz + y, x + z^2, x^2 + 2yz \rangle \cdot d\mathbf{r}$ from $(0, 0, 0)$ to $(1, 2, 3)$ using FTOLI.
(*Hint: find the potential function first.*)

Problem 5. Evaluate $\int_C (x^2 + y^2) ds$ where C is the helix $\mathbf{r}(t) = \langle \cos t, \sin t, t \rangle$, $0 \leq t \leq 2\pi$.